Learning Loss Technique for Active Learning on Image Segmentation

Farnaz Jazaeri **Niloofar Jazaeri** **Yang Xu** **Yuhong Guo**

Abstract

Even though there is abundant and easily accessible unlabeled data, the challenging and time-intensive task of labeling remains a hindrance to obtaining a satisfactory quantity of training data. This quantity is crucial for the development of effective supervised learning models. Active learning seeks to create algorithms that efficiently utilize labeled data by selectively querying the most informative samples for labeling by an oracle. The development of effective training approaches requiring fewer labels represents a significant research avenue. This direction is crucial for optimizing the utilization of computational and human resources in the labeling and training processes of deep neural networks. This research aims to enhance the training of an image segmentation model using active learning. Leveraging the learning loss technique, a secondary model predicts the loss of the primary model on unlabeled samples, guiding the annotation process by identifying instances where the model is uncertain. This innovative approach addresses the challenge of selecting the most informative samples for annotation, improving the efficiency and accuracy of the segmentation model. The empirical investigations conducted on the COCO benchmark dataset, including 5000 images, underscore the promising applicability of the loss-learning technique as an advanced uncertain query-based active learning model for image segmentation tasks. We validate our proposed method across diverse tasks including image classification, object detection, and human pose estimation, employing state-of-the-art network architectures. Additionally, our segmentation task serves as a novel validation for this approach, particularly within the realm of medical imaging, where the demand is extensive. Our optimized loss learning model is constructed with two fully connected layers and one softmax layer. Notably, we introduce an Unet-optimized model tailored for COCO image segmentation. The corresponding code is made publicly available.
Keywords: Loss Learning, Image segmentation, Active Learning, COCO dataset

1 Introduction

Active learning is an important subfield within machine learning, characterized by its approach to data annotation. The primary advantage of active learning lies in its ability to reduce the amount of data that needs to be annotated, while still maintaining or even improving the performance of the model. Unlike traditional machine learning, where a model is trained on a pre-labelled dataset, active learning involves an approach that iteratively selects those data instances that the model finds most uncertain or informative to be labelled; And by doing so, the model can learn more effectively from a smaller and more relevant dataset, which is particularly efficient in scenarios where labelling data is expensive, time-consuming or labour-intensive [1].

In practice, active learning involves a cycle where the model initially trains on a small set of labelled data. It then evaluates the unlabeled data and identifies instances that it's least certain about. These instances are then labelled by an expert or using existing annotation information and added to the

training dataset. This process repeats, allowing the model to continuously improve its performance while minimizing the need for extensive data labelling [2].

The segmentation of images is a crucial step in many computer vision applications. This precise extraction of relevant features from images enables applications that heavily rely on accurate and robust data interpretation. For instance, in medical imaging, accurate segmentation of tumours can directly impact diagnosis and treatment plans [3]. Similarly, in the field of autonomous driving, the ability to accurately segment and understand the surrounding environment can be the difference between a safe journey and an accident [4].

The application of active learning within segmentation tasks has emerged as a pivotal tool and holds immense promise due to its potential to streamline the annotation process for these intricate image analysis tasks. By intelligently selecting the most informative samples for labelling, active learning optimizes the segmentation model's training process, thus leading to improved accuracy and efficiency in segmenting complex visual data [5]. This targeted approach not only enhances the overall performance of segmentation models but also significantly reduces the human annotation effort, making it a crucial component in addressing the challenges of large-scale segmentation tasks.

This project aims to harness the power of the learning loss technique [6], leveraging another assisting secondary model to predict the loss of the primary model on unlabeled samples, thus enhancing the training of the loss values. This is particularly valuable because the loss function is indicative of how well the model is performing; lower loss generally means better performance. When the secondary model predicts high loss for certain images, it means that the primary model is unsure about them, and therefore, when these images are labelled, they can provide valuable information to improve the primary model. This innovative approach allows us to predict and effectively manage instances where the loss function increases on previously unlabeled data, addressing a significant challenge in selecting the most informative samples for annotation.

2 Related Work

Active Learning Techniques have recently been used to provide more labeled data for deep learning while the deep learning tasks became complex and expert annotation is an expensive resource to train Deep learning models with sufficient data samples [7].

Query Strategies: Active learning involves selecting the most informative samples for annotation to improve model performance with minimal labeled data. Popular query strategies include uncertainty sampling, query-by-committee, and information density-based methods [1]. State-of-the-Art Techniques: Recent advances in active learning include methods such as Bayesian Active Learning by Disagreement (BALD) [8], Core-Set Active Learning [9], and Deep Bayesian Active Learning with Image Data (BALD) [10]. Applications: Active learning has been successfully applied to various domains, including computer vision [9], natural language processing [11], and medical imaging [12].

Loss learning approaches focus on developing loss functions that better capture the model's performance. This can include custom loss functions tailored to specific tasks, such as segmentation or object detection. Instance Segmentation: Methods like Mask R-CNN8 and FCIS [13] have introduced loss functions designed for instance segmentation tasks, combining classification and segmentation losses. Adversarial loss, as seen in Generative Adversarial Networks (GANs) [14], has been employed to improve image generation tasks and has also influenced loss learning in other domains.

The landscape of active learning in the domain of loss learning methods has seen significant advancements, showcasing diverse strategies to enhance the efficiency and efficacy of model training through selective sample acquisition. DeSalvo et al. [15] introduce a pioneering active learning algorithm for streaming binary classification tasks, adeptly utilizing weak labels to minimize label requests. Their approach optimizes a surrogate loss on labeled and weak-labeled points, boasting both theoretical guarantees in the agnostic setting and superior empirical performance. Nguyen et al. [16] delve into adaptive pool-based active learning within a Bayesian framework, analyzing greedy active learning criteria. Wan et al. [17] contribute a novel unsupervised active learning approach, UALL, incorporating a deep neural network to model nonlinearity and considering representativeness, informativeness, and diversity. Yin et al. [18] propose a self-paced multi-criteria

active learning framework for image classification, addressing the deficiency of labeled data. Shukla and Ahmed [19] explore active learning for depth estimation using the learning loss method, conducting experiments that underscore its efficacy.

Additionally, Yoo and Kweon [6] present a task-agnostic active learning method leveraging a loss prediction module, enhancing model performance in image classification, object detection, and human pose estimation. In the realm of regression, Shukla and Ahmed [20] conduct a mathematical analysis of learning loss for active learning, proposing LearningLoss++ with rigorous gradient comparison and a novel convolutional architecture for enhanced model refinement. Long et al. [21] focus on active learning for ranking, introducing the Expected Loss Optimization framework for ranking functions, and demonstrating its potential through extensive experiments on real-world Web search datasets. Collectively, these contributions signify a rich tapestry of methodologies, each pushing the boundaries of active learning within diverse contexts and domains.

There are also Challenges and future work of active learning; Addressing Labeling Bias: Ongoing research focuses on addressing labeling bias in active learning by developing methods that consider the quality and diversity of annotations [14]. Active Learning in Federated Learning: The integration of active learning principles in federated learning scenarios is an emerging area, enabling more efficient and privacy-preserving model updates [22]. Benchmark Datasets for Active Learning: Efforts are being made to establish benchmark datasets for active learning evaluation, ensuring fair comparisons and fostering reproducibility [23].

3 Method

Algorithm 1: Learning Loss Technique for Active Learning

[24] **Data:** Labeled pool D_L, Unlabeled pool D_U, Labeling budget, Initialized models for θ_{loss}, θ_{seg}, θ_D, and θ_S
Result: Trained models θ_{loss}, θ_G, θ_D, and θ_S
$D_L \leftarrow$ Initial training with K samples with labels
until the labeling budget is finished
repeat
  for $e = 1$ *to epochs* **do**
    Sample $(x_L, y_L) \sim D_L$;
    Sample $x_U \sim D_U$;
    Compute $L_{seg} + \lambda L_{loss}$ using Eq. in [6];
    Update seg and $loss$ by descending stochastic gradient;
    $\theta_{seg} \leftarrow \theta_{seg} - \alpha_{seg} \nabla (L_{seg} + \lambda L_{loss})$;
    $\theta_{loss} \leftarrow \theta_{loss} - \alpha_E \nabla (L_{loss} + \lambda L_{seg})$;
    Compute L_{total} using Eq. 5;
    Update by descending stochastic gradient;
    $\theta_s \leftarrow \theta_{total_s} - \alpha_s \nabla L_{total_s}$;
  `// Pick K samples` X_s `from` D_U `with maximum segmentation module loss`
  $X_s \leftarrow \arg\max_{X_s} \text{Loss}(S(X_s))$;
  `// Label the selected samples by oracle, forming` (X_s, Y_o)
  Label the selected samples by oracle, which forms (X_s, Y_o);
  `// Update labeled and unlabeled datasets`
  $D_L \leftarrow D_L \cup (X_s, Y_o)$;
  $D_U \leftarrow D_U - X_s$;
until;
return Trained θ_{seg}, θ_{loss}, and θ_S;

The pseudo-code of the proposed algorithm, designed to overcome the challenges posed by the limited availability of labeled data, is presented in Algorithm 1. Since our goal is designed for segmentation algorithm the loss is picked based on the loss function of the segmentation model, Optimized UNet, The corresponding code is made publicly available at: `https://github.com/Neurotoxin0/COMP5116-Machine-Learning`.

3.1 Target Model

In the context of our semantic segmentation task, we employ the UNet model, originally introduced by Ronneberger et al. [25]. The nomenclature "UNet" derives from the distinctive U-shaped architecture it embodies. The core principle of UNet revolves around the aggregation of contextual information within the image through a two-stage process involving downsampling and subsequent decoding.

During the downsampling phase, the model applies multiple convolutional layers followed by max-pooling operations, effectively capturing and consolidating hierarchical features. The subsequent upsampling or decoding process aims to reconstruct the pixel-wise predictions by expanding the spatial dimensions. Notably, in this decoding process, each layer incorporates the feature map from the preceding layer, concatenated with skip connections.

The skip connections play a pivotal role in information transfer from the encoder to the decoder. Specifically, these connections facilitate the integration of high-level semantic information captured during the downsampling stage into the corresponding layers of the decoding phase. This mechanism enhances the model's ability to preserve fine-grained details and spatial information throughout the segmentation process.

This architectural design of UNet ensures an effective fusion of local and global contextual information, contributing to its success in semantic segmentation tasks. For a comprehensive understanding and technical details, readers are referred to the original work by Ronneberger et al. [25].

The UNet model's proficiency in semantic segmentation is noteworthy, but its utility for dense prediction tasks demands a significant volume of training data. This challenge is exacerbated in domains like medical imaging, where the necessity for expert annotation amplifies the cost and resource requirements of the training process. The intricate nature of semantic segmentation, coupled with the need for detailed labels, underscores the complexity of deploying UNet models. The involvement of domain experts is crucial, emphasizing the importance of skilled professionals in annotating the data. These factors collectively contribute to the resource-intensive nature of training, particularly in specialized domains. Researchers and practitioners must be mindful of these challenges, exploring potential solutions to alleviate data and annotation burdens.

3.2 Loss Module

The pivotal segment of our active learning system is the loss module, which is tasked with predicting loss values. Its primary function is to identify challenging samples in our extensive unlabeled pool where our main model, UNet, struggles to achieve precise segmentation.

In this module, feature maps are extracted from various layers of the UNet model. These maps undergo Global Average Pooling (GAP) followed by a Fully Connected layer (FC), with ReLU activation applied to each feature map extracted from the target model. Subsequently, the results are concatenated and fed into another Fully Connected layer to predict the loss value for the target model, as shown in Figure 1. This intricate process enables the identification of problematic instances, facilitating a targeted approach to active learning within the segmentation framework.

$$\text{feature}_1 = \text{RelU}(f_1(GPA(x_1))) \tag{1}$$

$$\text{feature}_2 = \text{RelU}(f_2(GPA((x_2))) \tag{2}$$

$$\text{feature}_3 = \text{RelU}(f_3(GPA(x_3))) \tag{3}$$

$$\text{predicted loss} = f_4(\text{feature}_1 \oplus \text{feature}_2 \oplus \text{feature}_3) \tag{4}$$

Where f_is are linear layers and x_is are feature maps from the UNet model.

3.3 Loss Function

To train the proposed model, a dual-model training approach is employed simultaneously. The first model is the UNet, trained akin to conventional segmentation models with a segmentation loss term L_{seg} measuring the dissimilarity between predicted ($\hat{y}$) and ground truth (y) segmentation outputs. The second model is the learning loss module, denoted as L_{loss}, responsible for predicting the loss values of the target model ($\hat{l}$). The comprehensive loss for training the entire system is a composite of the individual loss terms for both components. Specifically, it is defined as the sum of the segmentation loss (L_{seg}) and a scaled loss for the training loss module (L_{loss}), as expressed by the following formula:

$$L_{\text{total}} = L_{\text{seg}}(\hat{y}, y) + \lambda \cdot L_{\text{loss}}(\hat{l}, l) \tag{5}$$

Here, λ is a scaling constant that regulates the influence of the Loss module in the overall training process. This dual-model training strategy aims to jointly optimize both the UNet segmentation model and the Loss module for enhanced system performance.

3.3.1 Target model loss

The loss function employed for the UNet model is a combination of the standard cross-entropy loss, with predicted scores (P) and ground truth labels (G), and the Dice loss. The Dice loss is particularly valuable in segmentation tasks as it accounts for the spatial overlap between predicted and true segmentation masks. The cross-entropy loss captures the pixel-wise dissimilarity, while the Dice loss further refines the optimization by considering the intersection and union of the predicted and ground truth masks.

$$\text{Dice Loss} = 1 - \frac{2 \cdot \sum_i^N P_i \cdot G_i}{\sum_i^N P_i^2 + \sum_i^N G_i^2} \tag{6}$$

where N represents the total number of pixels in the segmentation masks. In the context of multi-class segmentation tasks, the Dice loss can be extended to accommodate multiple classes by averaging the individual class Dice coefficients.

3.3.2 Learning loss Module loss

For calculation (L_{loss}) in equation 5 we used the formula introduced in [6]. This function leverages pairs of samples within a mini-batch to compare the difference between predicted loss values, effectively discarding the overall scale of l. The Loss-prediction loss (L_{loss}) is defined as the maximum of 0 and the hinge loss between paired loss predictions, ensuring robust learning of the loss prediction module.

4 Experiment

Implementation Details: In our experimentation with the COCO dataset, we employed $K = 1000$ images in each cycle, systematically selecting and adding challenging samples to the training dataset. Each training cycle consisted of 50 epochs, with input images sized at $60 \times 40 \times 3$. We conduct training with specified system configurations, including an AMD 7950X3D CPU with 16 cores, 32 threads, two memory slots combing 64 GB of DDR5 memory, and an NVIDIA RTX 3090 GPU with 24 GB graphical memory and 10496 CUDA cores. Additionally, in most experiment we set λ hyperparameter in 5 to 1.

Model Performance: As depicted in Figure 3, our model demonstrates a significant performance advantage over all baselines across these datasets.

Evaluation Metrics: The selected metric proves highly effective, particularly when considering active learners with high computational costs that might encounter challenges in querying a sufficient number of points in real time. Notably, some active learners, such as the one presented in [26], demand substantial computational time even in low-dimensional spaces. We employ accuracy as the primary metric, a widely used measure, especially with balanced datasets [27]. For multiclass

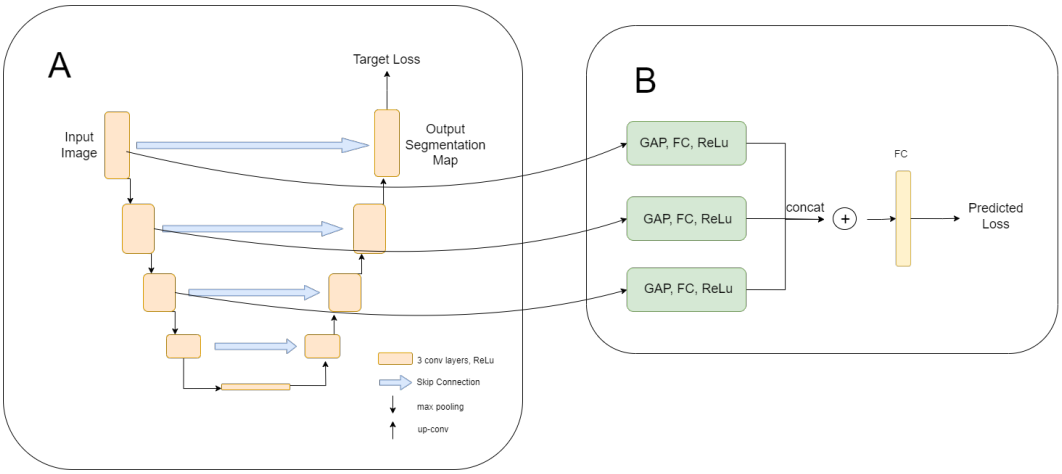

Figure 1: Illustrating the components of our active learning system for segmentation tasks. (A) showcases the schematic architecture of the UNet model, the cornerstone of our segmentation task. (B) highlights the loss module, leveraging feature maps generated at various levels of the UNet architecture to predict the loss value. This two-part framework underscores the synergy between the model structure and the learning loss process.

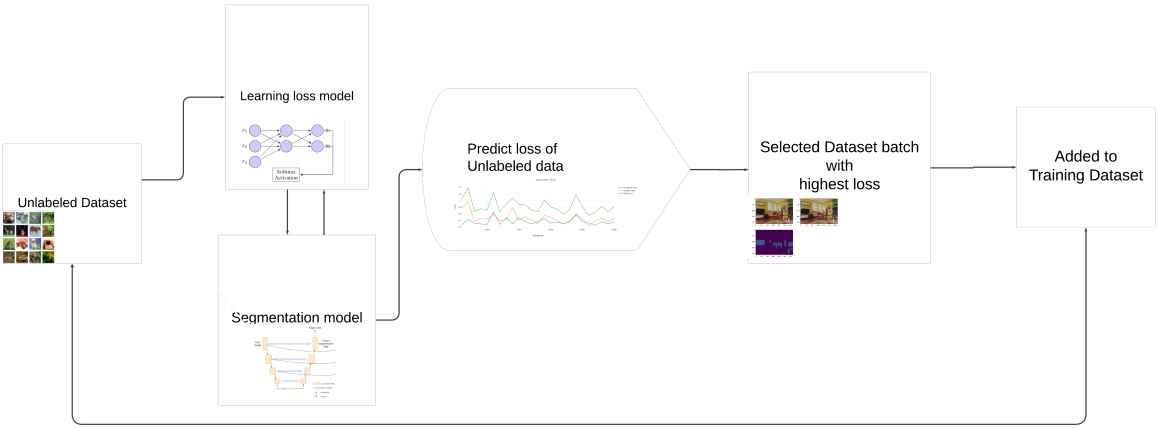

Figure 2: The pipeline of Active learning and the proposed method.

datasets, multiclass accuracy serves as a variant, representing the mean of the diagonal elements in the confusion matrix [28].

Architecture and hyper-parameters: In the realm of image segmentation, our approach centers on the utilization of the UNet architecture, incorporating nuanced architectural elements and carefully chosen hyper-parameters. The UNet design encompasses an encoder, strategically deployed to grasp contextual intricacies and construct a comprehensive feature representation. The bottleneck module, comprising convolutional layers, assumes a critical role in the preservation of spatial information.

Addressing the spatial resolution diminution inherent in the contracting path, the UNet architecture orchestrates a meticulous recovery process in the expanding path. A distinctive feature, the skip connections, establishes crucial links between corresponding layers in both the contracting and expanding phases. This intricate network of connections proves instrumental in safeguarding intricate details and gradient information during the intricate upsampling process.

Augmenting this, the skip connections coalesce with the output layer, endowed with a softmax activation function. The resulting output manifests as a segmentation mask, with each pixel intricately associated with a specific class, demarcating the likelihood of the pixel belonging to that designated category. Extending this architectural journey, the loss learning model is formulated with two fully

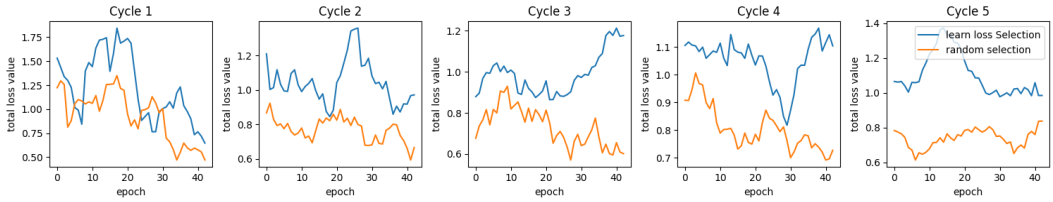

Figure 3: Dynamics of the training process captured through the evolution of loss values. The orange lines represent total loss under random sampling, contrasting with the blue curves that showcase total loss when strategically selecting and incorporating challenging samples based on the model's output. The curves are smoothed using a moving average with a window size of 7, providing a nuanced perspective on the model's learning behavior.

	Random Sampling			**Learn Loss sampling**		
Weight λ	**L seg**	**L loss**	**Total loss**	**L seg**	**L loss**	**Loss total**
0	0.028	-	0.202	**0.024**	-	0.235
0.5	0.034	0.031	0.189	0.093	0.148	0.167
1	0.070	**0.029**	**0.099**	0.035	0.218	0.254
1.5	0.069	0.493	0.143	0.144	0.11	0.209
2	**0.026**	0.136	0.298	0.093	**0.023**	**0.164**

Table 1: Minimum loss values under different conditions for different loss components.

connected layers culminating in a softmax layer, thereby contributing to a nuanced understanding and optimization of the segmentation process [29].

4.1 Dataset

The project mainly focuses on exploring the application of active learning in the field of image segmentation. To achieve this, the COCO (Common Objects in Context) dataset will be utilized, which is a widely used computer vision dataset that is known for its extensive annotations and diverse range of object categories [30].

The COCO dataset is maintained by a collaborative effort between Microsoft Research and the COCO Consortium, with many sponsors from leading companies and research groups like Microsoft, Facebook, CVDF, etc. [30]. And it was created to address the need for a large-scale dataset with comprehensive annotations for object detection, segmentation, and captioning tasks.

In this project, we utilized its official API for interfacing with the dataset as it offers many convenient features and less repeated coding work compared to raw use [31]. For the database version and category, version 2017 and the object segmentation category were utilized. It comprises all three sections (test, train, validate), with each containing 5,000 images. Specifically, 4,000 images were allocated for training, while 1,000 images were reserved for evaluation. The dataset spans 80 different classes, with pixel-wise annotations for segmentation, and by using the COCO API, the annotation information corresponding to each image can be directly accessed. Figure 5 and Figure 4 showcase samples from the dataset, featuring a mix of both simple and complex content.

4.2 Results

In Figure 3, we compare the dynamics of training the model under two different conditions: (1) when our system adds data to the training dataset, and (2) when data is randomly selected from the unlabeled pool, with respect to total loss values. In each cycle, the total loss for random sampling is consistently lower than the total loss for learning loss sampling, indicating that the proposed method selects samples that pose a greater challenge for the model in predicting segmentation. Moreover, the difference between the two curves increases as we progressively add more data in each cycle.

To observe a more evident dropping pattern in the learning curve, it is advisable to train the model for a greater number of epochs and add more samples in each cycle. This will provide a more comprehensive understanding of the model's performance over an extended training duration.

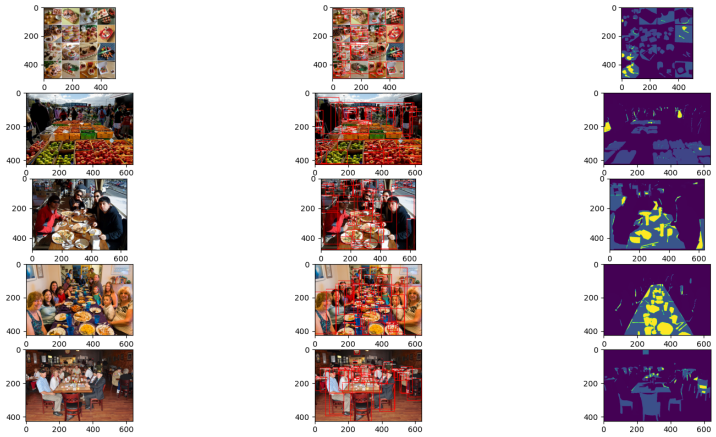

Figure 4: Five complex samples of dataset and the use of API for display the annotated version.

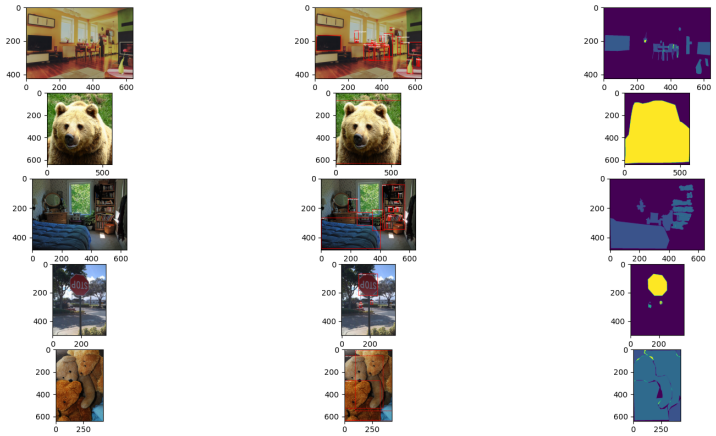

Figure 5: Five simple samples of dataset and the use of API for display the annotated version.

To evaluate the effect of different values of λ in Equation 5 on the training process, we trained the entire model with $\lambda \in \{0, 0.5, 1, 1.5, 2\}$ and report the minimum loss value during the training process in Table 1. According to the presented table, when we set $\lambda = 2$, both the total loss and L_{loss} exhibit the lowest values under the same condition.

5 Conclusion

The active learning approach offers a solution to achieve superior prediction accuracy at a reduced labeling cost, effort, and time. This is accomplished by identifying and querying the most informative and/or representative points among the unlabeled data points. In this study, we introduced a novel use of active learning strategy using the loss learning technique for segmentation tasks to build and be efficient. The deep learning structure tested for two types of distributed-based and the proposed uncertainty-based active learning algorithm and shows the superiority of the usage of the Loss learning technique which can be considered as a validation of improvement of previous works.

5.1 Limitations

The first limitation of this research is related to the inherent difficulty of the annotated segmentation task, especially in the intricate field of medical imaging - complex structures in medical images require expert-level annotations, which makes the training process challenging. Furthermore, despite utilizing learning loss techniques, further improvements are needed to optimize the loss prediction model. Such improvements are crucial for effective sample selection in active learning scenarios and for coping with the challenges associated with uncertain instances. Another limiting factor is that current segmentation benchmarks are sparse and may not fully cover the complexities encountered in the real world. Therefore, the development of more diverse and challenging benchmarks is essential to improve the robustness of segmentation models. In addition, the dataset used in this study lacks specificity for medical segmentation and autonomous driving applications: the lack of a dedicated dataset with a comprehensive collection of medical and automotive-related images limits the relevance of the model to these domains.

5.2 Future Study

Current work should focus on improving techniques to refine the loss function in segmentation tasks. Fine-tuning the learning loss model will enhance its ability to predict and select information samples, thus further improving the accuracy of the segmentation model. If possible, the development of more sophisticated benchmarks for segmentation tasks (especially medical imaging) is an area of interest: such benchmarks should be designed to reflect the complexities and challenges encountered in the real world. For the dataset component, future research efforts should prioritize exploring and identifying specialized datasets specifically for medical segmentation or autonomous driving applications, as datasets containing more relevant images will help improve model training. In addition, collaboration with domain experts can provide valuable insights and integrating expert feedback into the active learning process can also improve model performance and help to obtain more accurate segmentation results and increase utility.

References

[1] Burr Settles. Active Learning Literature Survey. Technical Report, University of Wisconsin-Madison Department of Computer Sciences, 2009. Accepted: 2012-03-15T17:23:56Z.

[2] Deep AI. Active Learning, May 2019.

[3] Geert Litjens, Thijs Kooi, Babak Ehteshami Bejnordi, Arnaud Arindra Adiyoso Setio, Francesco Ciompi, Mohsen Ghafoorian, Jeroen A.W.M. Van Der Laak, Bram Van Ginneken, and Clara I. Sánchez. A survey on deep learning in medical image analysis. *Medical Image Analysis*, 42:60–88, December 2017.

[4] Paulo R. Vieira, Pedro D. Félix, and Luis Macedo. Open-World Active Learning with Stacking Ensemble for Self-Driving Cars. 2021. Publisher: arXiv Version Number: 1.

[5] Sudhanshu Mittal, Joshua Niemeijer, Jörg P. Schäfer, and Thomas Brox. Best Practices in Active Learning for Semantic Segmentation. 2023. Publisher: arXiv Version Number: 2.

[6] Donggeun Yoo and In So Kweon. Learning Loss for Active Learning. 2019. Publisher: arXiv Version Number: 1.

[7] Alaa Tharwat and Wolfram Schenck. A Survey on Active Learning: State-of-the-Art, Practical Challenges and Research Directions. *Mathematics*, 11(4):820, February 2023.

[8] Choi Junhwan, Oh Seokmin, and Byun Joongmoo. Uncertainty estimation in AVO inversion using Bayesian dropout based deep learning. *Journal of Petroleum Science and Engineering*, 208:109288, January 2022.

[9] Ozan Sener and Silvio Savarese. Active Learning for Convolutional Neural Networks: A Core-Set Approach. 2017. Publisher: arXiv Version Number: 4.

[10] Neil Houlsby, Ferenc Huszár, Zoubin Ghahramani, and Máté Lengyel. Bayesian Active Learning for Classification and Preference Learning. 2011. Publisher: arXiv Version Number: 1.

[11] Vincent Vercruyssen, Lorenzo Perini, Wannes Meert, and Jesse Davis. Multi-domain Active Learning for Semi-supervised Anomaly Detection. In Massih-Reza Amini, Stéphane Canu, Asja Fischer, Tias Guns, Petra Kralj Novak, and Grigorios Tsoumakas, editors, *Machine Learning and Knowledge Discovery in Databases*, volume 13716, pages 485–501. Springer Nature Switzerland, Cham, 2023. Series Title: Lecture Notes in Computer Science.

[12] Xing Wu, Jie Pei, Cheng Chen, Yimin Zhu, Jianjia Wang, Quan Qian, Jian Zhang, Qun Sun, and Yike Guo. Federated Active Learning for Multicenter Collaborative Disease Diagnosis. *IEEE Transactions on Medical Imaging*, 42(7):2068–2080, July 2023.

[13] Yi Li, Haozhi Qi, Jifeng Dai, Xiangyang Ji, and Yichen Wei. Fully Convolutional Instance-Aware Semantic Segmentation. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4438–4446, Honolulu, HI, July 2017. IEEE.

[14] A. Kapoor, R.W. Picard, and Y. Ivanov. Probabilistic combination of multiple modalities to detect interest. In *Proceedings of the 17th International Conference on Pattern Recognition, 2004. ICPR 2004.*, pages 969–972 Vol.3, Cambridge, UK, 2004. IEEE.

[15] Giulia DeSalvo, Claudio Gentile, and Tobias Sommer Thune. Online active learning with surrogate loss functions. *Advances in Neural Information Processing Systems*, 34:22877–22889, 2021.

[16] Nguyen Viet Cuong, Wee Sun Lee, and Nan Ye. Near-optimal adaptive pool-based active learning with general loss. In *UAI*, pages 122–131. Citeseer, 2014.

[17] Chuanbing Wan, Fusheng Jin, Zhuang Qiao, Weiwei Zhang, and Ye Yuan. Unsupervised active learning with loss prediction. *Neural Computing and Applications*, pages 1–9, 2021.

[18] Tianxiang Yin, Ningzhong Liu, and Han Sun. Self-paced active learning for deep cnns via effective loss function. *Neurocomputing*, 424:1–8, 2021.

[19] Ilya Makarov and Ivan Guschenko-Cheverda. Learning loss for active learning in depth reconstruction problem. In *2021 IEEE 21st International Symposium on Computational Intelligence and Informatics (CINTI)*, pages 000115–000120. IEEE, 2021.

[20] Megh Shukla and Shuaib Ahmed. A mathematical analysis of learning loss for active learning in regression. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3320–3328, 2021.

[21] Bo Long, Olivier Chapelle, Ya Zhang, Yi Chang, Zhaohui Zheng, and Belle Tseng. Active learning for ranking through expected loss optimization. In *Proceedings of the 33rd international ACM SIGIR conference on Research and development in information retrieval*, pages 267–274, 2010.

[22] Reza Shokri and Vitaly Shmatikov. Privacy-Preserving Deep Learning. In *Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security*, pages 1310–1321, Denver Colorado USA, October 2015. ACM.

[23] Phil Ammirato, Patrick Poirson, Eunbyung Park, Jana Kosecka, and Alexander C. Berg. A Dataset for Developing and Benchmarking Active Vision. 2017. Publisher: arXiv Version Number: 2.

[24] Ali Mottaghi and Serena Yeung. Adversarial Representation Active Learning. 2019. Publisher: arXiv Version Number: 1.

[25] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional Networks for Biomedical Image Segmentation. 2015. Publisher: arXiv Version Number: 1.

[26] Alaa Tharwat and Wolfram Schenck. Balancing Exploration and Exploitation: A novel active learner for imbalanced data. *Knowledge-Based Systems*, 210:106500, December 2020.

[27] Rinu Boney and Alexander Ilin. Semi-Supervised and Active Few-Shot Learning with Prototypical Networks. 2017. Publisher: arXiv Version Number: 2.

[28] Christoph H. Lampert, Hannes Nickisch, and Stefan Harmeling. Attribute-Based Classification for Zero-Shot Visual Object Categorization. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 36(3):453–465, March 2014.

[29] Zongwei Zhou, Md Mahfuzur Rahman Siddiquee, Nima Tajbakhsh, and Jianming Liang. UNet++: A Nested U-Net Architecture for Medical Image Segmentation. In Danail Stoyanov, Zeike Taylor, Gustavo Carneiro, Tanveer Syeda-Mahmood, Anne Martel, Lena Maier-Hein, João Manuel R.S. Tavares, Andrew Bradley, João Paulo Papa, Vasileios Belagiannis, Jacinto C. Nascimento, Zhi Lu, Sailesh Conjeti, Mehdi Moradi, Hayit Greenspan, and Anant Madabhushi, editors, *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*, volume 11045, pages 3–11. Springer International Publishing, Cham, 2018. Series Title: Lecture Notes in Computer Science.

[30] Tsung-Yi Lin, Michael Maire, Serge Belongie, Lubomir Bourdev, Ross Girshick, James Hays, Pietro Perona, Deva Ramanan, C. Lawrence Zitnick, and Piotr Dollár. Microsoft COCO: Common Objects in Context. 2014. Publisher: arXiv Version Number: 3.

[31] cocodataset. cocodataset/cocoapi: COCO API - Dataset @ http://cocodataset.org/.